INCH-POUND

A-A-52054
September 29, 1989

COMMERCIAL ITEM DESCRIPTION

MIXING MACHINES, FOOD, ELECTRIC,

(VERTICAL, COMMERCIAL TYPE, BAKERY AND KITCHEN)

The General Services Administration has authorized the use of this commercial item description in preference to OO-M-38.

1. CLASSIFICATION.

1.1 Classification. This commercial item description covers food mixing machines of the sizes and types specified in table I.

1.2 Part or identifying number (PIN). A document based PIN to identify types and sizes of items is included in section 5.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The food mixing machines shall be of the vertical commercial type and shall be designed for mixing, blending, and whipping food ingredients using interchangeable agitators and whips. The machines shall be provided with a hub drive for operating with additional vegetable slicing and meat chopping attachments. The machines shall be capable of thoroughly mixing two full batches of bread dough without the insulated motor exceeding a temperature of 45°C and without damaging the attachments during operation. The time interval between batches must be thirty minutes or less. The recipe for the dough to be used for evaluation of mixer performance shall have an Absorbtion Ratio (AR) of 55 percent or more.

The Absorbion Ratio (AR) percentage is defined as:

$$\frac{\text{Weight of water}}{\text{Weight of dough}} = \text{AR \%}$$

The temperature of the dough shall be between 70 and 75 degrees Fahrenheit.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-ES Natick, MA 01760-5014.

AMSC N/A FSC 7320

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2.1.1 Mixing speeds. The machines shall be equipped with a three speed (minimum) selector for changing mixing speeds. The mixing speeds shall be: Low - 93 rpm to 139 rpm, Intermediate - 173 rpm to 285 rpm, High - 326 rpm to 591 rpm for mixing machines with three speeds. For mixing machines with four speeds, the speeds shall be: First - 40 rpm to 105 rpm, Second - 80 rpm to 200 rpm, Third - 165 rpm to 400 rpm, Fourth - 300 rpm to 600 rpm. The selector shall not move from one selected setting to another while in operation. A variable speed control, with clearly marked graduations, could be provided in lieu of a speed selector.

TABLE I. Mixing machines

Type	Size	Bowl Quantity	Model	Min. motor horse power	Electrical power source
1	5	1 ea. 5 qt	Bench	1/6	120 volts 60 Hz 1 phase
2	5	1 ea. 5 qt	Bench	1/6	220 volts 50 Hz 1 phase
3	5	1 ea. 5 qt	Bench	1/6	240 volts 50 Hz 1 phase
1	12	1 ea. 12 qt	Bench	1/4	120 volts 60 Hz 1 phase
2	12	1 ea. 12 qt	Bench	1/4	220 volts 50 Hz 1 phase
3	12	1 ea. 12 qt	Bench	1/4	240 volts 50 Hz 1 phase
1	20	1 ea. 20 & 12 qt	Bench	1/2	120 volts 60 Hz 1 phase
2	20	1 ea. 20 & 12 qt	Bench	1/2	220 volts 50 Hz 1 phase
3	20	1 ea. 20 & 12 qt	Bench	1/2	240 volts 50 Hz 1 phase
1	30	1 ea. 30 & 20 qt	Floor	3/4	120 volts 60 Hz 1 phase
2	30	1 ea. 30 & 20 qt	Floor	3/4	220 volts 50 Hz 1 phase
3	30	1 ea. 30 & 20 qt	Floor	3/4	240 volts 50 Hz 1 phase
1	60	1 ea. 60 & 30 qt	Floor	1.5	220 volts 50 Hz 1 phase
2	60	1 ea. 60 & 30 qt	Floor	1.5	240 volts 50 Hz 1 phase
3	60	1 ea. 60 & 30 qt	Floor	1.5	208 volts 60 Hz 3 phase
4	60	1 ea. 60 & 30 qt	Floor	1.5	380 volts 50 Hz 3 phase
1	80	1 ea. 80 & 40 qt	Floor	2	220 volts 50 Hz 1 phase
2	80	1 ea. 80 & 40 qt	Floor	2	240 volts 50 Hz 1 phase
3	80	1 ea. 80 & 40 qt	Floor	2	208 volts 60 Hz 3 phase
4	80	1 ea. 80 & 40 qt	Floor	2	380 volts 50 Hz 3 phase
1	140	1 ea. 140 & 80 qt	Floor	5	220 volts 50 Hz 1 phase
2	140	1 ea. 140 & 80 qt	Floor	5	240 volts 50 Hz 1 phase
3	140	1 ea. 140 & 80 qt	Floor	5	208 volts 60 Hz 3 phase
4	140	1 ea. 140 & 80 qt	Floor	5	380 volts 50 Hz 3 phase

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2.1.2 Mixing bowl support. The machine shall be equipped with an operating mechanism for raising and lowering the mixing bowl support. The adjustment mechanism for floor mounted machines shall be capable of raising full bowls from a dolly to the maximum operating height. Bowl supports shall have three lugs that securely connect the bowls to the mixers. The adjusting mechanism shall be self locking, or shall be provided with a positive lock to assure that moving components can be manually locked in position for safety. An automatic stop control shall be provided to prevent raising bowls beyond the maximum operating height.

2.1.3 Drive. The machines shall be driven by an electric motor connected through enclosed gearing, geared timing belt, V-belt(s), or enclosed chain driven mechanisms. Gearing, when used, shall not exceed 78 decibels A-scale for the bench mounted machines, and 85 decibels A-scale for the floor mounted machines while in operation (see 3.5). Cast unmachined gears are not acceptable. Electrical and power requirements are listed in 2.1.7. Metal guards shall be provided to protect the operator against contact with projecting or moving parts which are outside the main housing of the mixing machines. Plastic materials shall not be used for any guards or housings.

2.1.4 Housings. The vertical column, base, spindle support, and transmission housing shall be formed either as an integral unit or separate units and joined by welding, bolting, or a combination of both. The base shall be rigidly constructed to provide machine stability while in operation. The drive spindle shall be vertically mounted and securely held in place. Agitators and whips shall fit securely to the drive spindle and shall be easily attached and removed from the spindle without the use of tools.

2.1.5 Timer. The floor mounted mixing machines shall be provided with a 15 minute (minimum) timer. The timer shall automatically stop the machine in any speed after reaching the preset time. The timer design shall also include a nontiming position to permit continuous operation.

2.1.6 Starters. Bench mounted machines motors shall be fully enclosed, quick break, and shall have a manual starter switch. Floor mounted machines shall have either automatic reset starters or manual reset thermal overload and low voltage protection starters. Magnetic, thermal overload, automatic reset type switches shall be acceptable on mixers equipped with timers.

2.1.7 Electrical provisions. The bench mounted machines shall be provided with a 6 foot minimum electrical cord with attachment plug. The floor mounted machines shall have physical provisions for a permanent electrical connection. Unless otherwise specified (see 5.2), the electrical characteristics shall be as listed in table I.

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2.2 Accessories and attachments (see 5.2).

2.2.1 Mixing Bowls. The mixing bowls provided with each food mixing machine shall be within the capacity range listed in table I. Bowls shall be formed from a single piece of stainless steel. Stainless steel for mixing bowls shall conform to type 201 of ASTM A 412 or types 302 and 304 of ASTM A 167. Starting material thickness shall be nominal .0625 inch for bowls of 60 quart capacity and less and shall be nominal .0781 inch for bowls over the 60 quart capacity. Each bowl shall be equipped with two completely welded on handles. Spot welded handles are not acceptable. Each bowl shall have three lugs to match the lug positions of the bowl supports. The weld areas and deposited weld material shall be easily cleanable and corrosion-resistant.

2.2.2 Agitators. A batter beater, sweet dough beater, dough hook, pastry knife, wing whip, and wire whip suitable for use with each bowl size specified shall be provided with each food mixing machine. Beaters and hooks shall be fabricated of aluminum alloy or stainless steel. Wing and wire whips shall be made of stainless steel wire. Pastry knives shall be made of aluminum or stainless steel. Agitators shall be designed for cleaning without disassembly.

2.2.3 Colanders. A colander set shall be provided for the size 20 and 30 mixing machines. The colander set shall consist of a sieve holder, three sieves (fine, medium, and course), a roller beater or scraper beater, a drain bowl or mixing bowl, and necessary parts for assembling together and to the mixer. The sieve holder shall fit the drain bowl furnished as a component of the colander set or the size mixing bowl furnished with the mixer. The colander shall be adaptable for straining soups, making purees and sauces, and other similar work. The sieve holder and bowl shall be fabricated of stainless steel.

2.2.4 Vegetable slicer. When specified, a vegetable slicer capable of being attached to the mixer hub shall be provided. The attachment shall consist of knives enclosed in a housing and arranged to rotate in conjunction with a slicer plate. The attachment shall be capable of slicing, shredding, and grating vegetables, fruits, and nuts. The housing shall be fabricated of stainless steel or aluminum alloy and shall be provided with a hinged plate and operating handle to force the food to be sliced against the knives.

2.2.5 Slicer plate. The knife holder and slicer plate shall be of corrosion resisting metal. Knives shall be of corrosion-resisting cutlery steel, ground, and sharpened. The slicer plate for all sizes of food mixing machines shall not be less than 8-1/4 inches in diameter and knives shall be adjustable by external means from 0 to 1/2 inch with respect to the distance from the plate.

2.2.6 Accessories for the vegetable slicer. When specified, the vegetable slicer attachment shall be furnished with the following accessories, and shall be designed for installation in place of the slicer knife unit:

a. French fry or julienne cutter, suitable for cuts not less than 3/16 by 3/16 and not more than 1/2 by 1/2 inch.

b. 3/32 inch shredder plate.

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c. 3/16 inch shredder plate.

d. 5/16 inch shredder plate.

e. 1/2 inch shredder plate.

f. Extra fine grating plate.

g. Tubular or equivalent front for use with carrots, cucumbers, celery, or bananas.

2.2.7 Meat and food chopper. When specified, a meat and food chopper capable of being attached to the hub of the mixing machine shall be provided. The chopper shall be complete with manufacturer's standard pan and plunger.

2.2.8 Dollies. Unless otherwise specified, all floor mounted mixers shall be provided with a dolly capable of receiving, holding and rolling mixing bowls to and from their position at the machine without manually lifting the bowl and its contents. Dollies shall be fabricated of aluminum alloy.

2.3 Standards of compliance. The food mixing machines shall comply with applicable requirements of Underwriters Laboratories, Inc. (UL), Standard No. 763 and (NSF) National Sanitation Foundation (NSF) Standard No.8.

2.4 Data name plate (Government unique requirement (see 5.5). The food mixing machines shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data name plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to the food mixing machine by rivets, screws or welding in such a manner as to meet the applicable National Sanitation foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved or applied by photo sensitive means:

National Stock Number
Procurement Contract Number
Specification Data
Manufacturer's name, addaraess and telephone number
Supplier's name, address, and telephone number (list only if different from manufacturer)
Manufacturer's model number
Government approved manual number (when specified, see 5.2)

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Each name plate shall be placed so that it is visible to the operator during normal operating use and so as to not adversely affect the life and utility of the food mixing machine.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain subtantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and there after as may be otherwise provided for under the provisions of the contract.

3.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

3.1 End item examination. The end items shall be examined for defects listed in table II. The lot size shall be expressed in units of mixing machines of one type and size only. The sample unit shall be one mixing machine. The inspection level shall be II (see 5.3).

TABLE II. End item defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with document requirements	101	
Design	Any wearing part not accessible for adjustment and replacement		102
Construction and workmanship (applicable to all components and assemblies)	Any component missing or not as specified		103
	Any visible component fractured, split, or malformed		104
	Adjustable assembly that cannot be adjusted		105
	Adjustable assembly not properly adjusted to perform the function intended		106
	Casting miscast, cracked, porous, or contains blowholes		107
	Assembly or component improperly located (for example, gasket improperly fitted to provide seal)		108

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TABLE II. End item defects (cont'd)

Examine	Defect	Classification Major	Minor
Rivets	Rivet(s) missing or broken	109	
	Inadequately peened or loose	110	
Speed selector (gear driven)	Setting not positive at each position of gear selection, including neutral, if applicable	111	
	Marking missing, illegible, not conspicuously located, not permanently affixed, affected by disassembly procedure	112	
Lugs (bowl support and bowl)	Bowl support not provided with three lug positions	113	
	Bowl not provided with lugs to match the lug positions on the bowl support	114	
Stand (when specified)	Not open base welded construction	115	
	Top platform not reinforced	116	
	Edges of top and shelf not flanged or rolled down over frame and secured	117	
	Each leg not provided with an adjustable leveling device	118	
Welding	Missing, incomplete, through burn hole, cracked, severe undercut, or not fused	119	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		201
Accessories and attachments (as applicable)	Missing or not as specified	120	

3.4 Regulatory requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

3.5 Noise levels. Noise level tests shall be conducted in accordance with ANSI Standard No. S1.13 or certified to comply with the requirements specified in 2.1.3.

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4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export, as specified (see 5.2).

4.2 Commercial packaging and packing. Each mixing machine shall be cleaned, preserved, packaged, and packed in accordance with ASTM D 3951.

4.3 Export packaging and packing.

4.3.1 Export packaging. Each mixing machine shall be disassembled to the extent determined by the contractor to protect parts susceptible to damage during shipment and storage and for reduction of cubage. The bowls, agitators, accessories, and attachments shall be packaged in a domestic fiberboard box constructed and closed in accordance with PPP-B-636. The machine, bowls, agitators, accessories, and attachments shall be cleaned, preserved, cushioned, and immobilized to protect them against corrosion, deterioration, and physical damage during shipment and storage.

4.3.2 Export packing. Each complete mixing machine shall be packed in a snug fitting wood-cleated plywood box conforming to overseas type, style A or J, grade B, type 3 load of PPP-B-601 or nailed wooden box conforming to class 2, style 2 or 4, grade B, type 3 load of PPP-B-621. Each machine shall be securely bolted to the box with the bolts passing through the skids. The bolts shall not protrude below the skids. The machine, together with the fiberboard boxes within the wooden shipping container, shall be blocked and braced to prevent movement during transit. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-601 or PPP-B-621, as applicable.

4.4 Marking (commercial and export). Marking of shipping container shall be in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5. NOTES

5.1 Intended use. The food mixing machines are used in bakeries and kitchens for mixing doughs and blending and whipping food ingredients. The machines may be equipped with vegetable and meat preparing accessories.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type and size of mixing machine required (see 1.1).
c. Electrical requirements if other than specified (see 2.1.7).
d. Type and quantity of attachments required (see 2.2).
e. Government approved manual number to be included on the data name plate when manuals are purchased under the contract (see 2.4).
f. Selection of applicable packaging and packing required (see 4.1).
g. Acceptance criteria required (see 5.3).

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5.3 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

5.3.1 For end item visual examination. An acceptance quality level (AQL), expressed in terms of defects per hundred units of 2.5 for major defects and 6.5 for total (major and minor combined) defects is recommended.

5.4 Sources of documents.

5.4.1 Sources of government documents. Copies of Military and Federal documents are available from:

Naval Publications and Forms Center
ATTN: NPODS
5801 Tabor Avenue
Philadelphia, PA 19120-5099

5.4.2 Sources for nongovernment association documents.

ANSI Standard No. S1.13 - Methods for the Measurement of Sound Pressure Levels, is available from:

American National Standards Institute (ANSI)
1430 Broadway
New York, New York 10018

ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate Sheet, and Strip, and
ASTM A 4122 - Stainless and Heat-Resisting Chromium-Nickel-Manganese Steel Plate, Sheet, and Strip, and
ASTM D 3951 - Standard Practice for Commercial Packaging, are available from:

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

NSF Standard No. 8 - Commercial Powered Food Preparation Equipment, is available from:

National Sanitation Foundation (NSF)
3475 Plymouth Road
P.O. Box 1468
Ann Arbor, MI 48106.

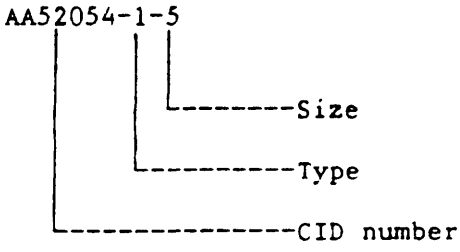
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UL Standard No. 763 - Motor Operated Commercial Food Preparation Equipment is available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062.

5.5 Government unique requirements. Whenever "(Government unique requirements)" is included in the title of a paragraph under "salient characteristics" it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

5.6 Part or Identifying Number (PIN). The PINs to be used for items acquired by this CID are created as follows:



MILITARY INTERESTS:

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review Activities:

Army - MD
Navy - MC, SA
DLA - GS
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

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